

Nabz نبض

A voice-first, Urdu-first AI health companion for Pakistani families. The browser captures speech and photos; a React SPA talks only to a FastAPI backend that owns every AI call, safety guardrail, and byte of patient data. Nabz runs an **adaptive clinical interview**, explains a non-diagnostic differential, routes to nearby care — and never states a diagnosis or invents a medicine.

Frontend React 18 • Vite • RTL • PWA

Backend FastAPI • Pydantic v2

AI Qwen + Qwen-VL • DashScope

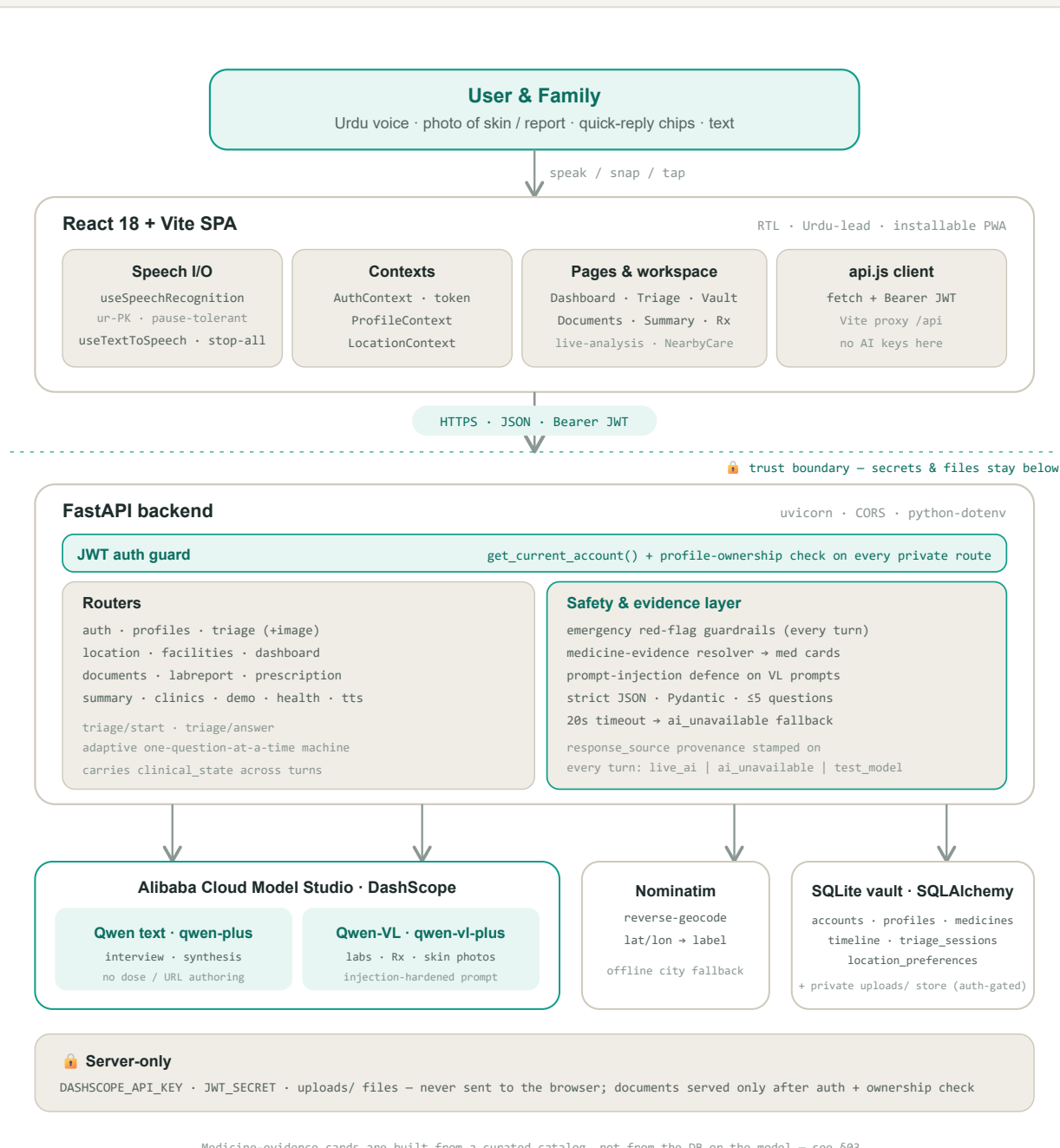
Data SQLite • SQLAlchemy

Care Location + ranked facilities

Auth JWT • bcrypt

01 The request path, top to bottom

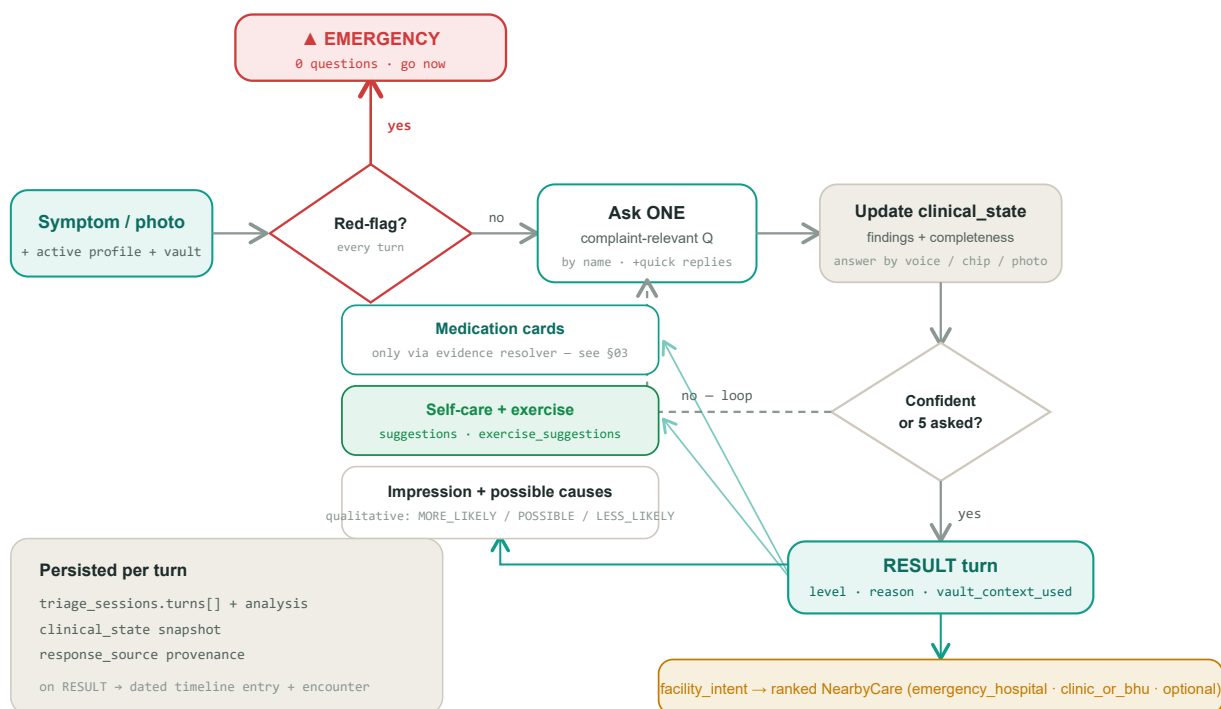
Every action flows down one spine: voice, photo or tap in the browser, through the React SPA, across a single JWT-authenticated JSON boundary, into FastAPI — which fans out to the Qwen models, a reverse-geocoder, the SQLite vault, and private file storage. AI keys and patient files live only below the boundary.



One JSON boundary, many sinks. The SPA holds a JWT but no model keys and never touches the file store. The router count grew — location, facilities, dashboard, documents, demo — but the boundary is unchanged: the browser still cannot reach DashScope, Nominatim, or a raw file directly.

02 Adaptive interview → clinical synthesis

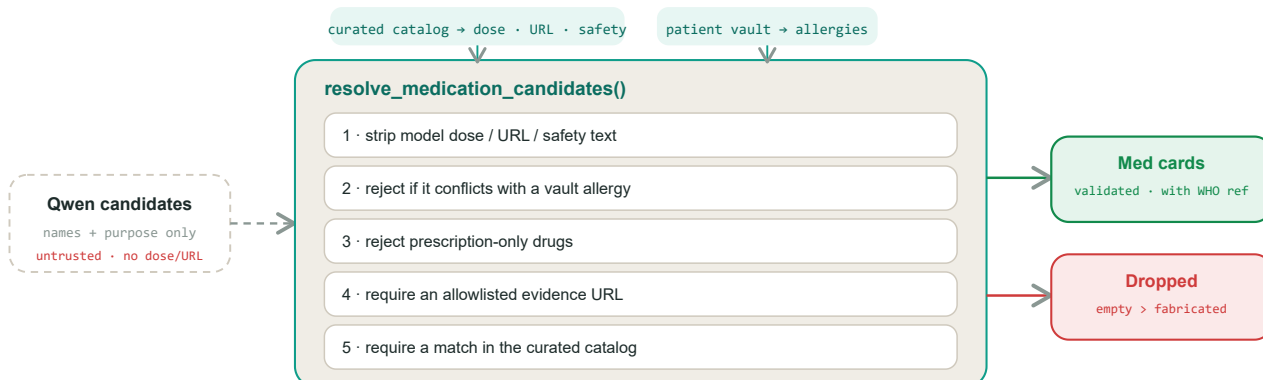
Nabz asks the single most useful *complaint-relevant* question at a time — a skin mark gets skin questions, not a reflex breathing question — accumulating a structured `clinical_state`. A red flag on any turn short-circuits to EMERGENCY. On conclusion it emits an explainable, non-diagnostic result.



The result is a briefing, not a verdict. Likelihoods stay qualitative on purpose — a chat or photo can't support a calibrated disease percentage. Medication cards never come straight from the model; they pass through the evidence resolver in §03. The red-flag check still runs on *every* turn, so danger reported late still escalates instantly.

03 The medication-evidence safety gate

This is why Nabz can suggest a medicine without prescribing one. The model only proposes *candidate names and purposes*; a server-side resolver is the sole writer of dose, evidence URL, and safety text — all drawn from a small curated catalog. Anything the model invents is dropped.



The model never authors clinical numbers. Dose, evidence URL, and safety text are read only from the curated catalog and the patient's own vault. The same catalog also annotates clinician-confirmed Vault medicines with WHO population-level context on the dashboard and doctor summary.

04 Location-first care routing

Location is a first-run input, not an afterthought — so every triage result can point to real, ranked facilities. The flow always answers, even on flaky Wi-Fi.

STEP 1

First-run gate

App withholds the workspace until a care location is set; the location screen stays editable later.

STEP 2

Resolve

/location/resolve reverse-geocodes lat/lon via Nominatim.

↳ offline → coarse PK-city lookup

STEP 3

Confirm

/location/confirm persists a LocationPreference (GPS or manual city).

STEP 4

Rank facilities

/facilities/nearby scores by severity-fit + type + real distance + verified + emergency capability.

05 Data model — the family vault

One account owns many patient profiles; everything hangs off a profile and is deleted with it (cascade). Personalization always pulls from the **active profile alone**. Location is stored per account; medicine evidence is derived at read time, never stored.

accounts

1 → ∞ profiles · 1 → 1 location

full_name, **phone** (unique)

password_hash · bcrypt

profiles

► medicines · timeline · sessions

display_name, relation, age, gender

blood_group, is_self

chronic_conditions[], allergies[], notes

medicines

∞ ◀ profile

name, strength, frequency, duration

with_food, notes

source manual | prescription

timeline_entries

∞ ◀ profile

kind triage | lab | prescription | document

title, subtitle, level

payload JSON · image_ref

triage_sessions

∞ ◀ profile

turns[], analysis, clinical_state

status, result_level, result_payload

response_source, initial_text

location_preferences

1 ◀ account

label, city, province

latitude, longitude, manual

accuracy_m, updated_at

uploads/ + documents

filesystem · timeline-linked

xray · mri · skin · lab · prescription

private store, auth + ownership gated

referenced by timeline payload

medicine evidence

derived · not stored

WHO context for confirmed meds

built from curated catalog

shown on dashboard + summary

06 Response provenance & demo seeding

The running app is **live-AI first**. Every turn is stamped with a `response_source` so a safe outage reply or a test double can never masquerade as a real assessment.

● Live path — `live_ai`

`DASHSCOPE_API_KEY` set

- qwen-plus adaptive interview + synthesis
- qwen-vl-plus for photos, labs, prescriptions
- Guardrails + evidence resolver run outside the model
- Timeout / bad output → safe fallback marked `ai_unavailable`

● Offline & demo

no key · tests · one-click demo

- pytest injects a `test_model` double — API contracts only
- Live clinical behaviour still needs a real key
- `/api/demo/hassan` seeds a synthetic longitudinal vault (mock-only, idempotent)
- 4 watermarked fixtures · confirmed Cetirizine · low-Hb CBC · ibuprofen allergy

07

API surface

All routes except auth, clinics, health and location-resolve require a bearer JWT; private routes also check profile ownership.

METHOD	ENDPOINT	PURPOSE	GUARD
Auth & profiles			
POST	/api/auth/register · /login	Create account (+ self profile) · issue JWT	public
...	/api/profiles · /{id}	Family profile CRUD (cascade delete)	jwt · owner
Triage & synthesis			
POST	/api/triage/start · /answer	Adaptive interview → explainable result	jwt
POST	/api/triage/.../image	Photo turn (skin / report) via Qwen-VL	jwt
Vault & documents			
GET	/api/dashboard/{id}	Patient-specific Home dashboard	jwt · owner
...	/api/documents · /{id}	Patient-scoped document vault (private files)	jwt · owner
POST	/api/labreport · /prescription · /confirm	Qwen-VL read → review → save to profile	jwt · owner
GET	/api/summary/{id}	Doctor handoff (+ docs + WHO refs)	jwt · owner
Location & care			
POST	/api/location/resolve · /confirm · /me	Reverse-geocode + persist care location	mixed
GET	/api/facilities/nearby	Ranked facilities by urgency + distance	jwt
Misc			
POST	/api/demo/hassan	One-click synthetic vault (mock-only)	jwt · mock

METHOD	ENDPOINT	PURPOSE	GUARD
GET	/api/clinics · /health · /tts	Sample facilities · status · spoken Urdu	public

Safety boundary. Nabz triages urgency, explains a non-diagnostic differential, and reads documents — it never states a confirmed diagnosis, never authors a dose, and never delays an emergency to keep asking. Suggested medicines pass the §03 evidence gate; prescription scanning only transcribes an existing clinician's paper and requires human confirmation before saving.